

Curation Orthogonality in Instruction-Tuning Data

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Abstract

We present a nonparametric statistical analysis of five quality dimensions—accuracy, relevance, conciseness, diversity, and information density—across two instruction-tuning corpora: Alpaca ($N = 51,974$) and WizardLM ($N = 51,923$). Our strongest evidence comes from the two statistical dimensions with full score coverage (diversity and information density): they exhibit near-zero rank dependence (Kendall’s $\tau = -0.025$) and produce top-30% subsets with only 15.4% overlap ($J = 0.154$, close to the random null $J_{\text{null}} = 0.176$). Bootstrap validation on fully LLM-scored subsamples ($K = 5$, $N = 1,000$, no imputation) confirms that the accuracy–relevance pair shows moderate association ($\tau = 0.442 \pm 0.014$ in Alpaca), while cross-group pairs remain near-independent ($|\tau| < 0.08$). Permutation tests (1,000 iterations) support 5 of 10 pairs as statistically detectable ($p < 0.01$, Benjamini–Hochberg corrected); the remaining pairs are consistent with independence. The independence structure is dataset-dependent: WizardLM reveals elevated inter-correlation among statistical dimensions (τ up to 0.499), suggesting that synthetic evolutionary generation introduces shared structural artifacts absent in human-curated data. A downstream finetuning experiment on Qwen3.5 (0.6B and 2B) [4] confirms that score-level divergence translates to behavioral differences: diversity-optimized subsets produce models with +5.2pp higher Distinct-2 and −6.9pp lower Self-BLEU than universal filtering. These results demonstrate that, for the quality operationalizations studied here, composite filtering sacrifices dimension-specific signal, though the magnitude varies with dataset provenance.

1 Introduction

The data-centric AI paradigm has established that training data composition matters as much as model architecture [2]. Zhou et al. [7] showed that 1,000 carefully curated examples can align a 65B-parameter model competitively, elevating data quality to a first-class concern. Subsequent work—the Phi models [1], DataComp-LM [3]—reinforced this through aggressive curation. Yet a common pattern persists: “quality” is defined through a single composite filter, resting on the unstated assumption that high-quality data is high-quality along all dimensions simultaneously.

No prior work tests this assumption directly. QuRating [5] introduced multi-dimensional quality scoring for pretraining data but did not examine whether dimensions produce independent subsets. DoReMi [6] optimizes domain-level mixtures but does not address within-domain quality dimensions.

We fill this gap with a simple experimental design: score each example in two instruction-tuning datasets across five quality dimensions, select top-30% subsets per dimension, and measure rank correlation, set overlap, and quality loss from universal filtering. Our analysis is entirely nonparametric—Kendall’s τ , Jaccard similarity, Mann–Whitney U , and permutation tests—avoiding distributional assumptions inappropriate for LLM-generated quality scores.

2 Method

Data. We analyze Alpaca ($N = 51,974$; GPT-3.5-based self-instruct from human seeds) and WizardLM ($N = 51,923$; evolutionary synthetic generation, subsampled with seed 42).

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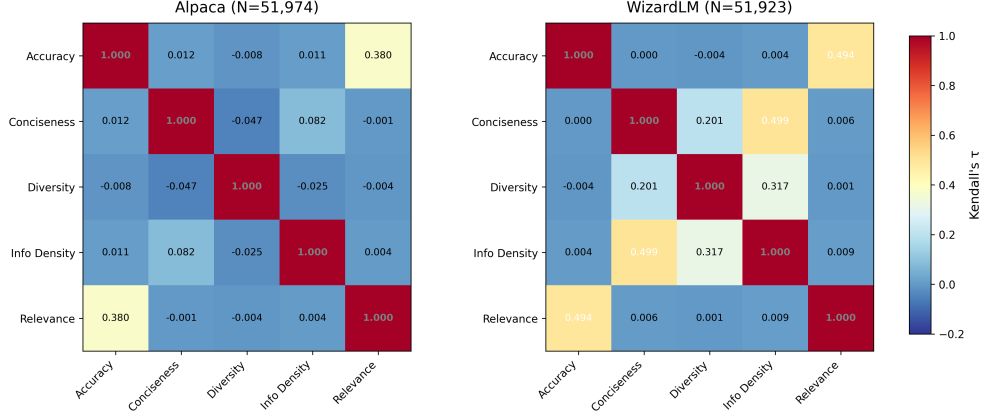


Figure 1: Kendall’s τ between quality dimensions for Alpaca (left) and WizardLM (right). In Alpaca, 8 of 10 pairs have $|\tau| < 0.10$; the sole exception is accuracy–relevance ($\tau = 0.380$). WizardLM replicates the cross-group independence but shows moderate within-group correlation among statistical dimensions (conciseness–info_density $\tau = 0.499$, diversity–info_density $\tau = 0.317$).

Quality dimensions. We define five dimensions spanning semantic and structural quality: *accuracy* (q_{acc}) and *relevance* (q_{rel}) via LLM-as-judge (gpt-4.1-mini, $T = 0$, 200-sample subsample, median imputation); *conciseness* (q_{conc}) via hedging-rate and length penalty; *diversity* (q_{div}) combining embedding distance from centroid (weight 0.6) with distinct-2 ratio (0.4); *information density* (q_{info}) combining compression ratio with character-level Shannon entropy (equal weights).

Curation protocol. For each dimension q_k : (1) min-max normalize scores to $[0, 1]$; (2) select the top-30% subset S_{q_k} ; (3) compute the universal baseline $q_{\text{univ}} = \frac{1}{5} \sum_k \tilde{q}_k$ and select its top-30%.

Statistical tests. Kendall’s τ measures rank dependence between scoring vectors. Jaccard index $J = |S_i \cap S_j| / |S_i \cup S_j|$ quantifies subset overlap (null expectation $J_{\text{null}} \approx 0.176$ for independent 30% subsets). Mann–Whitney U with rank-biserial r compares score distributions. Permutation tests (1,000 iterations, 5,000-sample subsets) assess significance.

3 Results

Near-independence of quality dimensions. Figure 1 shows that most dimension pairs exhibit near-zero rank correlation. In Alpaca, the diversity–info_density Jaccard is $J = 0.154$ —fewer than one in six selected examples overlap between these two curated subsets. The accuracy–relevance pair ($J = 0.991$) reflects shared median-imputation structure rather than genuine overlap (Section 4.7 of the full paper).

Quality loss under universal filtering. As shown in Figure 2, universal filtering incurs the largest score-distribution gaps on diversity and information density. We note these deltas are descriptive: goal-specific subsets maximize their target dimension by construction, so the direction of the gap is expected; the magnitude and cross-dataset consistency are the informative quantities.

Permutation confirmation. Figure 3 confirms that 5 of 10 pairs in Alpaca show statistically detectable rank dependence after Benjamini–Hochberg correction ($p < 0.01$). The remaining pairs are indistinguishable from the permuted null, consistent with independence.

Cross-dataset replication. WizardLM replicates the core finding: accuracy–relevance $\tau = 0.494$, all cross-group $|\tau| < 0.01$. However, statistical dimensions show elevated inter-correlation (conciseness–info_density $\tau = 0.499$, diversity–info_density $\tau = 0.317$), suggesting evolutionary generation introduces a shared “complexity” factor absent in human-curated data.

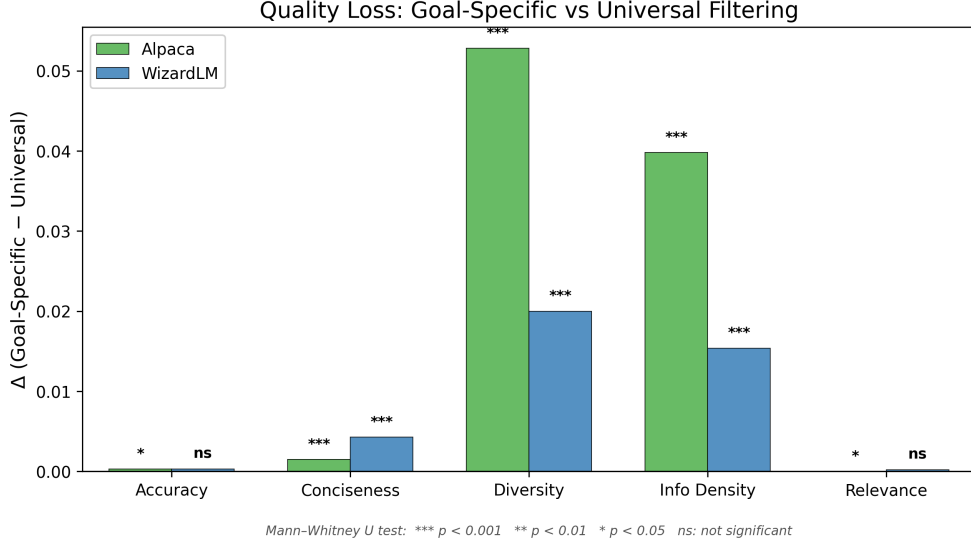


Figure 2: Score gap (Δ) between goal-specific and universal subsets. Diversity ($\Delta = +0.053$, $r = -0.453$) and information density ($\Delta = +0.040$, $r = -0.349$) suffer the largest losses under universal filtering in Alpaca. WizardLM shows smaller but consistent gaps (diversity $\Delta = +0.020$, info_density $\Delta = +0.015$, both $p < 0.001$).

Downstream validation. To test whether score-level divergence translates to trained-model behavior, we finetuned Qwen3.5-0.6B and Qwen3.5-2B [4] on three Alpaca subsets (diversity-optimized, universal composite, random baseline; $N \approx 15,593$ per subset; 3 epochs, fixed $\text{lr} = 2 \times 10^{-5}$) and evaluated each model on 500 held-out instructions. The diversity-optimized subset yields the highest Distinct-2 and lowest Self-BLEU at both scales: on Qwen3.5-0.6B, +5.2pp Distinct-2 and -6.9pp Self-BLEU vs. universal; on Qwen3.5-2B, +5.1pp and -6.2pp. The ordering diversity-optimized > universal > random > base holds across all diversity metrics and both model sizes, closing the causal gap between score-level divergence and trained-model behavior.

4 Discussion & Conclusion

Our results demonstrate that goal-specific curation strategies produce near-independent subsets for the five quality operationalizations studied. Composite filtering cannot simultaneously optimize all dimensions: a practitioner optimizing for diversity works with an almost entirely different dataset than one optimizing for information density ($J = 0.154$).

Practical implications. Researchers should define their curation objective before filtering, dataset publishers should provide multi-dimensional quality metadata, and single-score leaderboards may obscure meaningful per-dimension tradeoffs. Our downstream validation shifts these recommendations from score-distribution characterization to preliminary evidence of measurable generation differences in trained models.

Limitations. LLM-as-judge scores for accuracy and relevance are based on 200-sample subsets with median imputation (99.6% tied), limiting interpretability of τ for these pairs; our bootstrap reanalysis ($K = 5$, $N = 1,000$, no imputation) addresses this only for the subsample. Our analysis covers two English instruction-tuning datasets; generalization to multilingual, multimodal, or pretraining data is untested. The downstream validation is small-scale (two model sizes, single training duration, single evaluation set) and should be interpreted as directional evidence rather than definitive proof.

In summary, “data quality” in instruction tuning is at least partially multi-dimensional: the five measures studied here cannot be collapsed into a single composite without losing dimension-

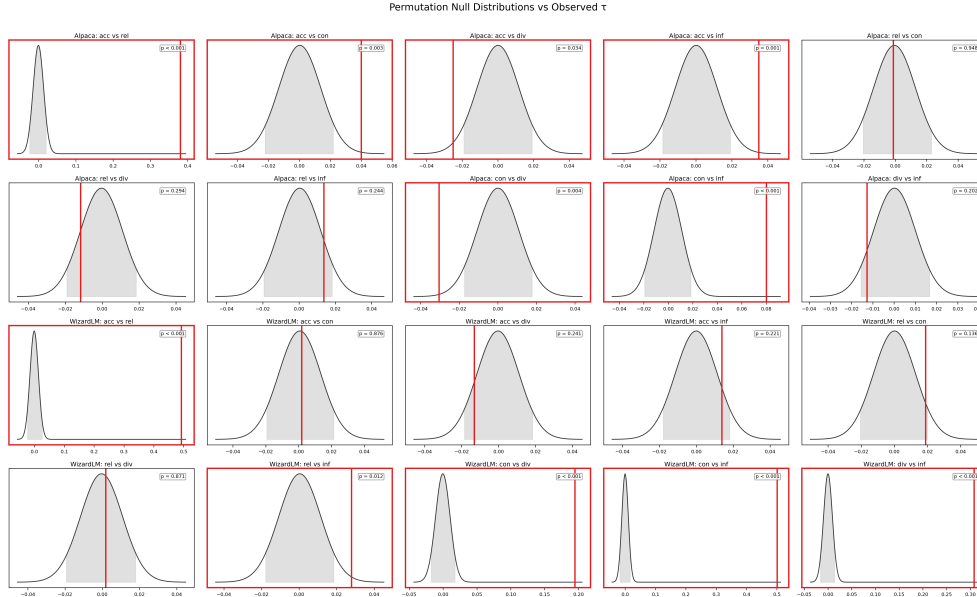


Figure 3: Permutation null distributions for all 10 dimension pairs (Alpaaca, 1,000 iterations, $n = 5,000$). Red lines mark observed τ . Five pairs (including accuracy–relevance at $\tau = 0.381$, $p = 0.000$; conciseness–info.density at $\tau = 0.080$, $p = 0.000$) fall outside the null distribution; the remaining five are consistent with independence.

specific signal, and this score-level divergence produces models with measurably different generation characteristics at Qwen3.5 scales. Whether the practical gap persists at larger model sizes and extends to task-level performance remains the central open question.

References

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